

Week 2 : Assignment 2

The due date for submitting this assignment has passed.Your last recorded submission was on 2026-08-03, 19:40 IST.

1. What are the disadvantages of the best fit partitioning algorithm?

- a) Might lead to fragmentation
- b) Entire process needs to be in RAM
- c) Limit the size of process by RAM size

Select the most suitable option from the following:

- ☒ a, b and c
- ☐ b and c
- ☐ a and c
- ☐ only a

Yes, the answer is correct. Score: 1

Accepted answer is a, b and c

2. Consider the following memory map using multiprogram with partition model. With reference to the given table, “first fit” will search from left to right. All sizes have the same unit (kB)

Size	65	175	150	125	150
Memory Status	In use	Free	In use	Free	In use

Which of the following order of memory requests will not be satisfied?

- ☐ 50, 80, 80 , following first fit
- ☐ 50, 80, 80 , following best fit
- ☒ 30, 80, 150 , following first fit
- ☐ 30, 80, 150 , following best fit

Yes, the answer is correct. Score: 1

Accepted answer is 30, 80, 150 , following first fit

3. What can you not find in a process's page table?

- ☐ Present bit
- ☐ Dirty bit
- ☐ Protection bits
- ☒ Sticky bit

Yes, the answer is correct. Score: 1

Accepted answer is Sticky bit

4. In a 32 bit processor, what is the maximum process size?

- ☐ 8GB
- ☐ 2GB
- ☒ 4GB
- ☐ 32GB

Yes, the answer is correct. Score: 1

Accepted answer is 4GB

5. What is the Dirty bit in the page table used for?

- ☐ Indicates if block is in RAM or not
- ☒ Indicates if contents of the RAM is modified with respect to its content in the swap space or the disk

Yes, the answer is correct. Score: 1

Accepted answer is Indicates if contents of the RAM is modified with respect to its content in the swap space or the disk

6. In 2 level page translation, given that the most significant 10 bits are used to index the page directory, if virtual address is of 32 bits and each page frame is of size 16 KB, then the number of entries in page table is

- ☐ 2^{10}
- ☐ 2^9
- ☒ 2^8
- ☐ 2^7

Yes, the answer is correct. Score: 1

Accepted answer is 2^8

7. In x86 systems, what converts logical address to linear address?

- ☐ CPU
- ☒ Segmentation unit
- ☐ Paging unit
- ☐ Physical Memory

Yes, the answer is correct. Score: 1

Accepted answer is Segmentation unit

8. In xv6, which of the following are correct (multiple options might be correct) ? V-> virtual P-> physical

- ☒ $V2P(a) (((uint) (a) - KERNBASE)$
- ☐ $P2V(a) (((uint) (a) - KERNBASE)$
- ☐ $V2P(a) (((void *) (a) + KERNBASE)$
- ☒ $P2V(a) (((void *) (a) + KERNBASE)$

Yes, the answer is correct. Score: 1

Accepted answer is $V2P(a) (((uint) (a) - KERNBASE)$, $P2V(a) (((void *) (a) + KERNBASE)$

9. Which function in xv6 removes free pages from the linked list's head?

- ☐ kfree()
- ☒ kalloc()
- ☐ malloc()
- ☐ free()

Yes, the answer is correct. Score: 1

Accepted answer is kalloc()

10. When a CPU boots, which of the following registers are not set to zero (multiple options might be correct)?

- ☒ Code segment register
- ☒ Instruction pointer
- ☐ Stack pointer
- ☐ Base pointer

Yes, the answer is correct. Score: 1

Accepted answer is Code segment register, Instruction pointer